AMERICAN CONTINENTAL RAILROAD INC. / ACRI

TIME, CONNECTIVITY AND THE NEXT AMERICAN RAIL SPINE

Economic prospectus for a national high-speed network with a 300-400 km/h design ambition and a staged hydrogen-electric propulsion programme.

300-400
km/h target speed band

6
regional systems proposed

2055
planning horizon context

CONCEPT-STAGE STRATEGIC PROSPECTUS | 11 OCTOBER 2026

Prepared for economists, infrastructure investors and public-sector decision-makers. Not an offer of securities; all corridor times and network benefits are illustrative, not investment forecasts.

INVESTMENT THESIS

A national network is worth more than a collection of fast lines

ACRI's opportunity is to build a connected set of high-speed regional systems - each linking dense economic nodes, airports, universities, advanced-manufacturing clusters and state capitals - then connect those systems into a national spine. A 300-400 km/h design ambition can materially shorten journeys in the 150-600 mile market, where rail can compete with driving and selected short-haul flights.

300-400
km/h maximum design ambition; not average end-to-end speed

230-250
km/h working average used for core corridor examples

~$0.82bn
annual time value in central illustrative volume case

THE ECONOMIC CASE

Core proposition: cut generalized door-to-door travel time between regional economic nodes, expand the effective labour market, and make multi-city business travel possible without routine car or airport transfers.

Four findings for economists

- Connectivity compounds. When Chicago, Dallas, Atlanta, Denver, Los Angeles and the Northeast Corridor work as transfer nodes, more origin-destination pairs become viable without building a dedicated direct route for every pair.
- Time is measurable; wider productivity is not automatic. Value travel-time savings using USDOT parameters, then test labour-market access and agglomeration as separate effects. Avoid double-counting wages, induced trips or station-area property uplift.
- Deliver by corridor, not by national megaproject. Start with routes where city pairs are close enough for rail to beat the car door-to-door, and build interchange capacity so each initial route has network value.
- Hydrogen is a strategic technology bet. Hydrogen-electric trains at 300-400 km/h are not a mature, commercially proven U.S. operating template. ACRI should treat propulsion validation and low-carbon hydrogen supply as formal stage gates, not as guaranteed advantages.

What this prospectus is - and is not

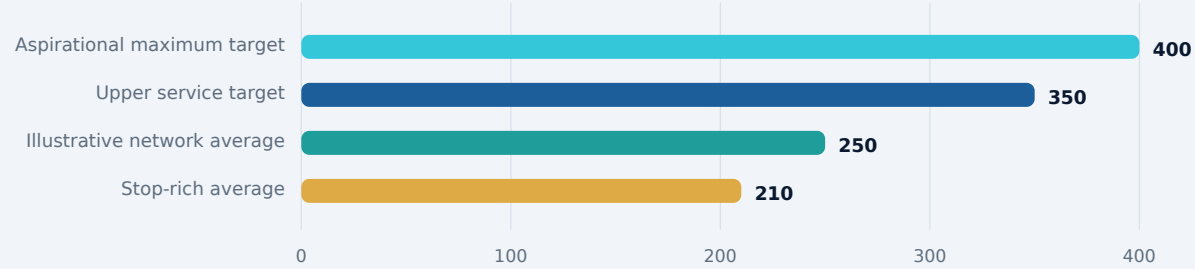
This is a concept-stage economic and network prospectus, not a feasibility study, bankable business plan or securities offering. Timetable examples are modelled from approximate corridor lengths and assumed average speeds. They are not official alignments, fare forecasts, environmental determinations, ridership forecasts or cost estimates. Capital economics must be validated corridor by corridor.

01 / THE TIME-SAVING MECHANISM

The right metric is door-to-door time, not headline speed

A 400 km/h train does not average 400 km/h from city centre to city centre. Acceleration, deceleration, stops, curves, station approaches, pathing, terminal access and timetable recovery all reduce commercial average speed. This prospectus uses 230-250 km/h as a planning average for selected express examples, with 10-15 minutes added for terminal and dwell allowances. Short, stop-rich corridors may average less.

DESIGN SPEED IS NOT AVERAGE JOURNEY SPEED



km/h; schematic planning values, not an operating timetable

Indicative calculation

Input	Planning assumption	Why it matters
Track distance	Illustrative corridor miles converted to kilometres	Must be replaced with surveyed, engineered alignment distance.
Commercial average	230-250 km/h on selected express runs	Reflects acceleration, curves and some intermediate dwell; not the peak speed.
Service allowance	10-15 minutes	Allows for terminal approach, dwell and timetable margin in a simplified model.
Comparator	Approximate road journey time, normally without major congestion	Actual savings depend on traffic, parking, station access, frequency and fares.

CORRIDOR SCREENING RULE

A useful early screen: prioritise city pairs where a high-speed service can make the main intercity segment roughly 1-3 hours and maintain useful frequency. A fast track with poor station access or infrequent trains may produce weak door-to-door savings.

Speed bands: 300 km/h ≈ 186 mph; 350 km/h ≈ 217 mph; 400 km/h ≈ 249 mph. A true 400 km/h passenger system requires dedicated, fully grade-separated infrastructure, stringent alignment geometry, signalling and safety systems, and a rolling-stock platform engineered for that operating environment. The FRA's historic national-rail framework similarly treats core express as dedicated infrastructure for large urban areas, though the exact U.S. network and standards must be updated and assessed corridor by corridor. [1][2]

02 / ILLUSTRATIVE CITY-PAIR ECONOMICS

Where the time savings could be largest

The examples below are an initial market-screening exercise rather than a promise of future timetables. Distances are approximate corridor lengths for planning discussion; road times are rounded baseline estimates and exclude live traffic variation. The high-speed column uses an assumed 230-250 km/h average plus terminal/dwell allowance; detailed stop patterns can shift results materially.

Illustrative corridor	Route mi	Road baseline	ACRI HSR model	Approx. saving
Chicago - St. Louis	~284	~4 h 45	**~2 h 00**	**~2 h 45**
Chicago - Detroit	~283	~4 h 20	**~2 h 00**	**~2 h 20**
Chicago - Minneapolis-St. Paul	~410	~6 h 15	**~2 h 50**	**~3 h 25**
Dallas - Houston	~239	~3 h 45	**~1 h 45**	**~2 h 00**
Dallas - Austin	~195	~3 h 05	**~1 h 25**	**~1 h 40**
Houston - San Antonio	~197	~3 h 00	**~1 h 30**	**~1 h 30**
San Francisco - Los Angeles	~383	~6 h 00	**~2 h 40**	**~3 h 20**
Atlanta - Charlotte	~245	~4 h 05	**~1 h 45**	**~2 h 20**
Seattle - Portland	~174	~2 h 45	**~1 h 25**	**~1 h 20**
New York - Washington, DC	~225	~4 h 30	**~1 h 35**	**~2 h 55**

How to read the table

These numbers illustrate the potential order of magnitude, not the predicted commercial timetable. New York-Washington is particularly sensitive to comparator choice because the corridor already has frequent intercity rail; the road comparison is shown only for consistency, while formal benefit-cost work should compare high-speed rail against the actual no-build travel options for each market. Similarly, express patterns with very few stops can be faster but may reduce the number of communities served.

WHAT THE TIME SAVING ENABLES

A 2-hour saving on a 250-mile trip can change the geography of work: same-day client meetings, split-site teams, university collaboration and airport-to-downtown access become easier. It does not mean each saved hour turns directly into GDP; economists should start with the welfare value of time and test wider impacts separately.

Planning implication: route priority should be ranked on generalised journey time, market size, frequency, interchange opportunities, construction complexity and delivered cost per passenger-hour saved - not maximum speed alone.

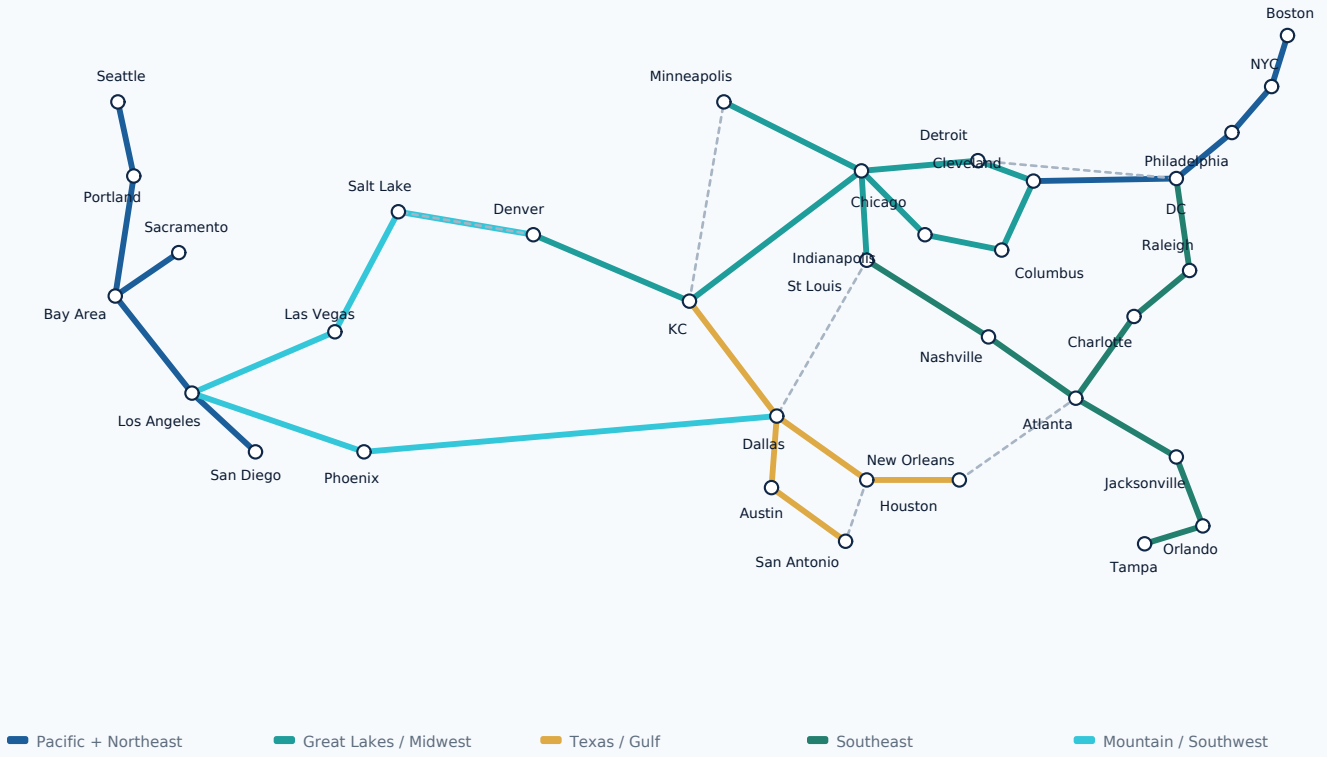
03 / A NATIONAL NETWORK ARCHITECTURE

Regional systems first; national reach through transfer nodes

ACRI should organise delivery around regional systems with common ticketing, timed connections, consistent service standards and a small number of high-capacity interchange hubs. The schematic below is a network concept, not an alignment proposal; several long-distance links require extensive route, demand and cost studies.

ACRI CONCEPTUAL NATIONAL SPINE

schematic only - not geographically accurate



Network layer	Role in the national system	Node logic
Regional express	Dense travel between nearby metros; high frequency and reliable centre-city stations.	Two or more major cities within a competitive 1-3 hour rail market.
Interregional trunk	Connect regional systems; provide through services or low-friction transfers.	Hubs such as Chicago, Dallas, Atlanta, Denver and Los Angeles.
Feeder and access	Bring secondary cities, university towns and airports into the express system.	Regional rail, transit, coach and airport interchanges; do not force every train onto HSR track.

Why hubs matter economically

A well-connected hub increases the number of reachable destinations for any given infrastructure mile. A city can gain access to multiple regions through one good interchange; ACRI can therefore target network effects earlier than a point-to-point national buildout. The design objective is low transfer penalty, high frequency, clear wayfinding, integrated ticketing and robust recovery from delay. FRA regional-planning guidance explicitly treats rail as part of an integrated system with transit, aviation, highways and non-motorised access. [2]

04 / POTENTIAL DOMESTIC NODES

Six regional systems with distinct economic roles

Node selection should reflect population and employment catchments, trip generation, land-use potential, airport access, university and medical networks, freight-industry clusters, construction feasibility and the value of connecting onward journeys. The initial list below is deliberately selective; it is a concept for screening, not a final route map.

A. Great Lakes & Midwest

Candidate nodes: Chicago; Milwaukee; Minneapolis-St. Paul; Detroit; Cleveland; Columbus; Indianapolis; St. Louis; Kansas City; Omaha.

Chicago is the strongest candidate for a central interchange because it sits at the heart of established rail geography and the FRA has already prioritised corridors to Minneapolis-St. Paul, St. Louis, Indianapolis and Detroit. Connect manufacturing, finance, health, higher education and regional business markets. [3]

B. Texas & Gulf

Candidate nodes: Dallas-Fort Worth; Austin; San Antonio; Houston; New Orleans as a later extension.

A compact Texas triangle offers multiple sizeable city pairs before long interregional links are attempted. Dallas and Houston can become north-south/east-west transfer nodes; airport connectivity and station placement are decisive. Gulf extensions need hurricane resilience and careful demand testing.

C. Northeast & Great Lakes arc

Candidate nodes: Boston; New York; Philadelphia; Washington, DC; Pittsburgh; Cleveland; Detroit.

This region has the strongest existing intercity rail market and can demonstrate high-frequency integration. Incremental capacity, reliability and bottleneck relief may create more value than treating every segment as a fresh 400 km/h new-build line.

D. Southeast Crescent

Candidate nodes: Washington, DC; Richmond; Raleigh-Durham; Charlotte; Atlanta; Jacksonville; Orlando; Tampa.

A chain of intermediate markets lets the system serve both intercity and metropolitan travel. The Southeast Regional Rail Plan offers an existing multistate planning base; HSR should complement rather than displace useful regional services. [2]

E. Pacific Coast

Candidate nodes: Seattle; Portland; San Francisco Bay Area; Sacramento; Los Angeles; San Diego.

High-demand coastal city pairs, airport links and major technology, entertainment, research and trade centres support the concept. California’s 494-mile Phase 1 is planned to connect San Francisco and Greater Los Angeles in under three hours, providing a useful benchmark for market scale, not a guarantee of delivery. [4]

F. Mountain & Southwest

Candidate nodes: Los Angeles; Las Vegas; Phoenix; Salt Lake City; Denver; Kansas City as an eastbound bridge.

Build where topography and city-pair demand support credible corridors. Las Vegas-Los Angeles, the Front Range and Phoenix connections have different engineering profiles; desert distances and water, heat, grades and elevation make route-specific modelling essential.

Node investment test

For each station, ACRI should publish catchment size, station-to-CBD time, local-transit links, interchange penalty, expected weekday/off-peak service, accessibility, housing and employment pipeline, and cost per forecast passenger. Avoid prestige stations that are isolated from the urban fabric.

05 / ECONOMIC VALUATION

Monetise time first; treat productivity spillovers carefully

USDOT's FY2026 Benefit-Cost Analysis Guidance provides recommended travel-time values in 2024 dollars and instructs applicants to document demand, the no-build baseline, costs and benefit timing. The general all-purpose travel-time value used here is $21.80 per person-hour. For high-speed rail competitive with air, USDOT specifies higher values for personal and business travel; ACRI's corridor BCAs should use the appropriate mix only when supported by market evidence. [1]

ILLUSTRATIVE ANNUAL VALUE OF TRAVEL-TIME SAVINGS



Annual trips x hours saved x $21.80/person-hour (USDOT FY2026 guidance; 2024 dollars).

Illustrative case	Annual affected trips	Average hours saved	Person-hours saved	Gross annual time value
Low	10 million	1.0	10 million	**$218 million**
Central	25 million	1.5	37.5 million	**$817.5 million**
High	50 million	2.0	100 million	**$2.18 billion**

Calculation: affected one-way trips per year x average hours saved x $21.80 per person-hour. These scenarios are sensitivity cases, not a passenger forecast, revenue projection or benefit-cost ratio. They exclude induced-trip benefits, transfer and waiting-time benefits, operating cost savings, road-safety impacts, emissions, construction costs and operations/maintenance costs. A final BCA should add only benefits supported by evidence and prevent double counting.

Wider economic channels to test

Channel	Potential transmission mechanism	Measurement discipline
Labour-market reach	More jobs become reachable within a tolerable commute or same-day trip.	Use accessibility metrics and observed travel behaviour; distinguish commuters from business trips.
Business productivity	More face-to-face meetings, easier multi-site work and thicker supplier/customer markets.	Use firm data or credible evaluation; avoid counting wages and monetised time savings twice.
Station-area development	Housing, office, retail and hotel investment around accessible stations.	Measure net additional value after displacement, land costs, public subsidy and local infrastructure.
Safety and road costs	Mode shift may reduce driving exposure, congestion and vehicle operating costs.	Apply route-specific diversion rates, occupancy and USDOT parameters.

USDOT's illustrative general travel-time values are $20.10/person-hour for personal travel, $34.60 for business travel and $21.80 for all-purpose travel; air-competitive high-speed rail values are $53.50 personal and $86.00 business. Values are in 2024 dollars; apply the actual trip-purpose mix and correct BCA conventions. [1]

06 / HYDROGEN TRACTION: DEVELOPMENT GATE

Make hydrogen a validated platform, not an unpriced assumption

Hydrogen-electric traction could support energy diversification and reduce tailpipe emissions, but hydrogen does not by itself make a train high-speed or zero-carbon. The 300-400 km/h ambition is primarily a rolling-stock, alignment, signalling, aerodynamics, braking, power and infrastructure challenge. At this speed range, the traction architecture must demonstrate sufficient continuous power, acceleration, thermal management, onboard energy storage, crashworthiness and operational range without consuming passenger capacity or compromising safety.

TECHNOLOGY STATUS

Critical diligence finding: hydrogen fuel-cell passenger trains have been demonstrated in regional service, with published top speeds around 140-160 km/h on current examples; a commercially proven 300-400 km/h hydrogen trainset and U.S. network operating model should not be assumed. ACRI must finance a staged technology programme and maintain an electrified high-speed fallback.

Recommended propulsion strategy

Workstream	ACRI position	Pass / fail evidence
300 km/h demonstrator	Fuel-cell electric drive plus high-power battery buffer; pursue a production engineering partner and test on controlled dedicated track.	Verified high-speed stability, braking, energy consumption, range, crashworthiness, refuelling turnaround and fleet availability.
350-400 km/h target	Treat as a separate development gate, not an automatic extension of the 300 km/h design. Study liquid-hydrogen storage and mass/volume implications.	Independent safety case, aerodynamic / pantograph-equivalent system validation, traction-power test, route compatibility and lifecycle cost.
High-speed trunk infrastructure	Retain a 25 kV AC overhead-electric baseline for cost, energy and technical-risk comparison; design rolling stock and network interfaces for interoperability.	Whole-system lifecycle model comparing renewable electricity, hydrogen supply, conversion losses, capex, maintenance and resilience.
Hydrogen hubs	Site production or contracted supply at depots and major hubs; storage, compression/liquefaction, dispensing, emergency response and redundancy.	Delivered $/kg, hourly throughput, supply contracts, safety approvals and contingency stocks.
Carbon integrity	Prefer verified low-carbon hydrogen; report well-to-wheel emissions, electricity source and leakage assumptions.	Transparent carbon-intensity accounting; do not market train emissions as zero without lifecycle evidence.

Hydrogen supply-chain reality

DOE identifies delivery infrastructure such as pipelines, liquefaction facilities, trucks, storage, compressors and dispensers as part of the hydrogen system. DOE workshop work also highlights cryogenic infrastructure, testing, validation and hydrogen losses as unresolved deployment needs for heavy-duty transport. A high-speed fleet with concentrated refuelling demand could help anchor an industrial hydrogen hub, but the infrastructure costs and utilisation rates must be modelled with the railway rather than assumed away. Current Alstom and Siemens regional hydrogen train examples publish top speeds of 140 km/h and 160 km/h respectively, underscoring the size of the engineering step to 300-400 km/h. [5][6][9]

Decision rule: if high-speed hydrogen fails the performance or lifecycle-cost gate, ACRI should use electrified high-speed trunk operations and reserve hydrogen for non-electrified regional feeders, depot equipment or other uses where it earns its place economically.

07 / INVESTMENT MODEL

ACRI should invest through staged corridor concessions

A national passenger system is unlikely to be financeable from one undifferentiated revenue forecast. ACRI should create separate corridor-level project companies or concessions with transparent asset, demand, operating and funding accounts, while retaining national standards for rolling stock, ticketing, digital signalling, customer experience, maintenance and procurement.

Potential revenue and value-capture lines

Revenue line	Commercial logic	What to validate
Passenger fares	Core recurring revenue; fare yield varies by distance, peak, class, frequency and competing modes.	Origin-destination demand, elasticity, occupancy, yield management and the effect of fares on accessibility.
Station development	Leases, retail, offices, hotels, parking and joint development near well-connected stations.	Land ownership, zoning, delivery timing, market absorption and additionality.
Operations & maintenance	Long-term maintenance concessions, rolling-stock servicing, track systems and digital operations.	Fleet size, renewal profile, availability regimes and whole-life cost.
Hydrogen supply / offtake	Supply agreements or joint ventures at selected depots, if the rail fleet creates firm demand.	Production pathway, hourly throughput, take-or-pay exposure, price indexation and lifecycle carbon.
Public availability / capital support	May be needed for socially valuable network benefits not fully recoverable through fares.	Competitive BCA, public-value case, risk allocation and affordability for each state/federal partner.

Capital-market design principles

- Separate infrastructure and train operations where it improves transparency; define access charges and performance standards contractually.
- Use project finance only after alignment, land, environmental review, demand, utility relocations and delivery risk have reached bankable maturity.
- Standardise designs and procurements across corridors to reduce the cost of bespoke viaducts, stations, signalling and rolling stock, but do not force a uniform alignment where geography differs.
- Allocate demand risk to the party best able to manage it. Early corridors may need public capital, federal loans or grants, tax-advantaged bonds and station-area development participation.
- Publish a corridor-level financial model and separate it from the economic BCA: fares are revenue to ACRI but largely a transfer in social cost-benefit accounting.

CAPITAL DISCIPLINE

No defensible national capex figure is offered here. U.S. high-speed construction cost varies sharply with tunnelling, viaducts, property, utility relocation, urban approaches, local delivery conditions and scope. The investment case must begin with corridor-specific reference designs and cost ranges, not a single national cost-per-mile extrapolation.

Public-sector framework: USDOT’s 2026 BCA guidance recommends a 7% real discount rate for federal discretionary grant analyses (unless applicable programme guidance says otherwise). Use the required public BCA alongside an equity cash-flow model, sensitivity cases and affordability analysis. [1]

08 / DELIVERY ROADMAP & RISK

Prove the corridor, prove the train, then scale the network

Stage	Illustrative timing	Work to complete	Gate to proceed
1. Screen	2026-2028	OD demand, corridor mapping, node catchments, no-build travel time, environmental fatal-flaw screening, initial public/private partner structure.	Shortlist corridors on passenger-hour value, affordability, delivery complexity and network contribution.
2. Engineer & demonstrate	2028-2031	Reference design; independent cost ranges; hydrogen supply feasibility; 300 km/h prototype programme; risk and safety case.	No full procurement before traction, range, crashworthiness, refuelling and lifecycle cost are independently validated.
3. Build a first regional system	Early 2030s onward	One high-value corridor or connected pair; stations integrated with transit and airports; robust service and operating data.	Actual construction performance, ridership, reliability, fare yield and maintenance cost meet published thresholds.
4. Add interregional trunks	After first-system proof	Connect successful regional systems using common ticketing, timetable architecture, control systems and vehicle standards.	Incremental network value is positive after accounting for transfer penalties and connection risk.
5. Optimise nationally	Long-run horizon	Add feeder cities, second stations where justified, service frequency and resilient alternative routing.	Continuous BCA and public reporting; no route extension by political logic alone.

Risk register - the issues that can break the business case

Risk	Why it matters	Mitigation / proof point
Construction and land	Costs and schedules can overwhelm the value of time.	Independent quantity survey, route alternatives, early utility and property work, risk-adjusted contingency.
Hydrogen power density / storage	Tank mass and volume, traction power, range and refuelling may be incompatible with a 400 km/h commercial timetable.	Prototype with a third-party safety case; compare an electrified reference train on a whole-system basis.
Ridership / fares	Low volumes or fare resistance can undermine cash flows.	OD-level demand model; ticket/frequency sensitivity; phased capacity; transparent demand risk allocation.
Network coordination	Late connections and poor station access erode door-to-door value.	Integrated timetable, cross-platform transfers where possible, through-ticketing, disruption recovery and local transit agreements.
Energy / carbon	Hydrogen can shift emissions upstream if produced from carbon-intensive sources; delivered costs may vary.	Contract low-carbon supply, verify lifecycle intensity and stress-test production, transport and power prices.
Governance / cross-state	Fragmented approvals and funding delay coherent regional delivery.	Multistate compact, common standards, corridor authorities, published milestones and clear federal-state-private obligations.

The public record establishes planning precedents, not a ready-made business case. The FRA’s Midwest framework extends to 2055 and spans 12 states; the Southeast has its own multistate planning process. ACRI’s commercial task is to turn those regional plans into financeable, interoperable projects with credible cost and performance evidence. [2][3]

09 / ECONOMISTS' SCORECARD

What ACRI should prove before it asks for scale capital

For each proposed corridor, ACRI should publish a consistent scorecard. This allows investors to compare projects and lets policymakers distinguish the social case from the private cash case.

Measure	Required analysis	Investment / policy question
Door-to-door time	No-build and build time including access, wait, transfer, reliability and frequency.	Does rail win on generalized time for a meaningful share of trips?
Passenger demand	OD matrix by trip purpose, daypart and competing mode; induced travel separately.	Are forecast loads and fares credible under low/base/high cases?
Time-value benefits	Apply correct USDOT values, occupancy and user categories; discount by year.	What is the gross benefit per year and present value of time saved?
Capital and lifecycle cost	Design-stage capex range, risk contingency, renewals, energy and operations.	Can the corridor be delivered affordably and maintained for decades?
BCR / NPV	Real discount rate, analysis period, residual value and all material benefits/costs.	Does the public economic case clear the relevant programme threshold?
Private finance	Fares and ancillary revenue, debt service, equity return, downside and refinancing.	Who takes demand, construction, energy and technology risk?
Hydrogen readiness	Prototype performance, supply-chain reliability, carbon intensity and delivered fuel cost.	Is hydrogen superior to or justified against overhead electrification on this corridor?
Network effect	Incremental OD pairs and interchange value versus isolated route.	Does this project strengthen the national network or merely add mileage?

BOTTOM LINE

ACRI recommendation: advance a 300 km/h-class high-speed corridor programme with a disciplined hydrogen-electric demonstrator, but make the national rail system independent of any unproven propulsion promise. Build the first routes where route geometry, market demand, urban access and public BCA align; use the evidence to decide whether 350-400 km/h and hydrogen-powered trunk operation justify further capital.

Investor diligence list

- Independent route options and cost range for each proposed first-wave corridor.
- Passenger demand and fare model benchmarked against driving, existing rail, air and intercity coach.
- A hydrogen-versus-overhead-electric whole-life model, including fuel production and delivery infrastructure.
- A signed governance and delivery framework across participating states, cities, rail operators and utilities.
- Separate economic BCA, project-finance model and equity return case; no use of headline time savings as revenue.

SOURCES, ASSUMPTIONS & DEFINITIONS

Evidence base and modelling limits

This prospectus draws on public U.S. transportation and energy sources. Corridor and timetable examples are authored scenario calculations, not reported official forecasts. Sources checked for this concept prospectus on 11 October 2026.

[1] U.S. Department of Transportation, Benefit-Cost Analysis Guidance for Discretionary Grant Programs, 2026 Update (December 2025). Establishes recommended travel-time values, BCA structure and a 7% real discount rate for federal discretionary grant analyses. Table A-2 gives values in 2024 dollars and identifies elevated values for air-competitive high-speed rail.

Open USDOT 2026 BCA guidance

[2] Federal Railroad Administration, Regional Rail Planning. Describes regional planning purpose and integrated rail links with local transit, aviation, highways and non-motorised modes; includes the Midwest and Southeast multistate studies.

Open FRA regional rail planning

[3] Federal Railroad Administration, Midwest Regional Rail Plan release and final report. The release identifies a 12-state planning area, a 2055 framework, Chicago as a major regional rail hub and cornerstone corridors to Minneapolis-St. Paul, St. Louis, Indianapolis and Detroit.

Open FRA Midwest plan overview · Final report PDF

[4] California High-Speed Rail Authority, Project Overview. States Phase 1 at 494 miles with a planned trip between San Francisco and Greater Los Angeles of under three hours, with additional extensions identified in Phase 2.

Open California HSR project overview

[5] U.S. Department of Energy, Hydrogen Delivery. Describes required delivery infrastructure including pipelines, liquefaction, trucks, storage, compressors and dispensers, and the need for infrastructure expansion as demand grows.

Open DOE hydrogen delivery

[6] U.S. Department of Energy, Hydrogen and Fuel Cell Technologies Office, Hydrogen Infrastructure Priorities to Enable Deployment in the High-Impact Transportation Sector (2024 workshop). Identifies cryogenic infrastructure, testing, validation and losses among key issues for heavy-duty transport deployment.

Open DOE hydrogen infrastructure workshop

[7] Amtrak, FY25 Year-End Ridership Fact Sheet (November 2025). Reports 34.47 million total riders in FY2025, including 15.17 million NEC Spine riders; included as demand context, not as forecast for ACRI.

Open Amtrak FY25 ridership sheet

[8] U.S. Department of Transportation, Brightline West bond authority announcement (23 January 2024). Gives context for a U.S. high-speed rail project planned at 218 miles, capable of 186 mph or more, with a cited project value of $12 billion in that announcement. It is not a direct capital-cost benchmark for ACRI.

Open USDOT Brightline West announcement

[9] Manufacturer technical information on existing regional hydrogen train platforms: Alstom Coradia iLint states a maximum speed of 140 km/h; Siemens Mireo Plus H states a top speed of 160 km/h. These are regional vehicles, not direct high-speed train comparators.

Alstom Coradia iLint · Siemens Mireo Plus H

Definitions and key assumptions

Design / peak speed: Highest intended operating speed under appropriate track and train conditions. Commercial average speed: Distance divided by in-vehicle elapsed time, including the effect of stops and acceleration. Door-to-door time: Includes access, wait, transfer and egress time as applicable. Person-hours saved: affected one-way trips multiplied by average hours saved. Gross time value: person-hours saved multiplied by the selected value of time; it is a welfare measure, not company revenue.

Hydrogen qualification: the report does not claim that 300-400 km/h hydrogen-powered passenger trains are currently commercially proven. Hydrogen’s lifecycle emissions depend on production pathway, electricity input and delivery. ACRI should compare hydrogen-electric and overhead-electric systems on a full lifecycle basis before committing to a fleet standard.

Prepared as a strategic prospectus for economists and infrastructure decision-makers. ACRI corridor locations, timings, demand and costs remain subject to feasibility studies, environmental review, engineering design, consultation and independent due diligence.